

PROPOSAL FOR TENDER AND FUNDING

Project Title: Construction of Gabions for Riverbank Stabilization and Flood Mitigation along the Ngong River

Submitted by:

Andolo Gideon Youths

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Submitted to:

The County Executive Committee Member (CECM) for Environment, Water, and Natural Resources,

[Insert Name of County, e.g., Nairobi City County Government],

P.O. Box [Insert Box Number],

[Insert City].

Date: [Insert Date]

1.0 EXECUTIVE SUMMARY

Andolo Gideon Youths (AGY) seeks to partner with the County Government to address the critical issue of riverbank erosion and seasonal flooding along the Ngong River. This proposal outlines a comprehensive plan for the construction of durable gabion structures. The project aims to protect infrastructure, prevent property loss, restore environmental integrity, and empower local youth through employment and skills development. We request to be considered for tender and funding to execute this vital community infrastructure project.

2.0 ORGANIZATION INFORMATION

2.1 Name of Organization

Andolo Gideon Youths (AGY)

2.2 Mission and Vision

- **Vision:** To be a premier community-based organization fostering sustainable environmental solutions and empowering youth for a resilient and prosperous community.
- **Mission:** To engage, train, and deploy local youth in environmental conservation and infrastructure projects, promoting stewardship, economic empowerment, and community safety.

2.3 Track Record

Andolo Gideon Youths is a registered community-based organization with a proven track record of successful community engagement. While our primary focus has been on environmental clean-ups and tree-planting initiatives within the informal settlements along the Ngong River, our members possess significant practical experience. Our key strengths include:

- **Community Trust:** We have established strong working relationships with local leaders, residents, and businesses in the riparian areas, ensuring seamless community mobilization and project acceptance.
- **Labor Force:** We command a dedicated and disciplined workforce of over 50 trained youth members ready to be deployed for manual labor, site security, and community liaison roles.
- **Past Projects:** Successfully coordinated five community-led clean-up drives along a 2-km stretch of the Ngong River, removing over 10 tons of solid waste and plastic debris. We have also partnered with local environmental NGOs on tree-planting and awareness campaigns.

3.0 PROJECT JUSTIFICATION

3.1 Problem Statement

The Ngong River is facing severe environmental stress, with uncontrolled erosion being a primary threat. During the rainy seasons, the river swells, and the unstable, denuded banks collapse, leading to:

- **Flooding:** Flash floods inundate homes, businesses, and roads, displacing families and disrupting economic activities.
- **Property Damage:** The erosion undercuts the foundations of houses, pathways, and utility lines, leading to costly damage and posing significant safety risks.
- **Environmental Degradation:** Erosion increases siltation and turbidity in the water, degrading aquatic habitats and polluting the river with debris and soil runoff. The proliferation of solid waste trapped in eroded banks exacerbates the problem.
- **Safety Hazards:** Collapsing banks create dangerous, unstable zones, particularly hazardous for children and pedestrians living near the river.

3.2 Proposed Solution

The construction of gabions—rectangular wire mesh cages filled with durable stone—presents the most effective and environmentally appropriate solution for this context. Gabions will:

- **Stabilize Banks:** The mass of the stone-filled cages will resist the hydraulic forces of the river, preventing further bank collapse and lateral erosion.
- **Absorb Energy:** The flexible, permeable structure dissipates the energy of flowing water, reducing scouring at the riverbed and toe of the bank.
- **Integrate with Environment:** Unlike solid concrete walls, gabions allow for natural drainage and can eventually become habitats for vegetation, integrating seamlessly into the riparian ecosystem.

3.3 Benefits

- **Flood Mitigation:** Reinforced banks will contain the river within its channel during peak flows, drastically reducing flood risk for adjacent communities.
- **Infrastructure Protection:** Stabilizing the banks will safeguard roads, drainage lines, and residential structures.
- **Enhanced Environmental Quality:** Reduced erosion will improve water quality. The project will also include a component for clearing solid waste from the targeted sections before construction, leaving the river cleaner.
- **Community Empowerment:** This project will directly employ local youth, providing them with income and valuable skills in construction, stone masonry, and project management.

4.0 PROJECT SCOPE AND METHODOLOGY

4.1 Location

The project will be implemented along critical, high-priority sections of the Ngong River, specifically identified in collaboration with the county's water resources authority. The proposed initial phases are:

- **Section A:** From [Landmark, e.g., the footbridge near Kibera Police Post] to [Landmark, e.g., the Masaba Road crossing] (Approx. 300 meters).
- **Section B:** A 150-meter stretch near [Landmark, e.g., the Lang'ata Road Bridge] identified as a high-erosion hotspot.

4.2 Gabion Design

We propose using standard, high-durability gabion baskets:

- **Type:** Rectangular, welded or woven mesh gabions.
- **Material:** Galvanized steel wire with a PVC coating for enhanced corrosion resistance.
- **Dimensions:** Standard basket size of **2m (L) x 1m (W) x 1m (H)**. These will be stacked in a stepped configuration to match the slope of the riverbank.

- **Filling:** Durable, angular, non-porous stones (e.g., rip-rap or machine-crushed stone) ranging from 150mm to 300mm in diameter.

4.3 Construction Process

1. **Site Preparation & Mobilization:** Conduct a site survey, clear vegetation, remove solid waste, and establish access routes. Erect safety signage.
2. **Excavation & Foundation Preparation:** Excavate a level foundation trench at the base of the slope. Lay a geotextile filter fabric to prevent soil migration and ensure proper drainage.
3. **Gabion Assembly:** Assemble the wire mesh cages on-site, linking them together using galvanized ties to form a continuous, monolithic structure.
4. **Stone Filling:** Place stones in the cages manually or with a small excavator, ensuring proper compaction and minimal void space. Larger stones will be placed on the outer face for aesthetic and structural integrity. Bracing rods will be used to prevent bulging during filling.
5. **Lacing and Capping:** Once filled, the lids will be securely laced shut. Successive layers will be offset (stepped back) and firmly tied to the layer below.
6. **Quality Control:** Continuous supervision will ensure correct assembly, stone placement, and tie spacing to meet engineering standards.
7. **Rehabilitation:** The area behind the gabions will be backfilled with suitable soil, and native grass or shrubs will be planted to further stabilize the area and improve aesthetics.

4.4 Timeline (6-Month Project Schedule)

Activity	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6
Mobilization & Site Preparation						
Material Procurement & Delivery						

Activity	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6
Gabion Assembly & Installation (Section A)		■	■			
Gabion Assembly & Installation (Section B)				■	■	
Backfilling & Revegetation					■	■
Quality Control & Handover						■

5.0 RESOURCE REQUIREMENTS

5.1 Materials

Item	Description	Quantity	Unit Cost (KSh)	Total Cost (KSh)
Galvanized Steel Gabion Cages	2m x 1m x 1m, PVC coated	500 units	8,500	4,250,000
Stones/Rip-Rap	Machine-cut, durable stone (150-300mm)	2,500 m ³	3,500	8,750,000
Geotextile Fabric	Woven, high-tensile filter fabric	3,500 m ²	250	875,000
Galvanized Ties & Lacing Wire	Heavy-duty wire for assembly	500 kg	300	150,000

Item	Description	Quantity	Unit Cost (KSh)	Total Cost (KSh)
Bracing Rods	Steel rods to prevent bulging	2,500 units	200	500,000
Subtotal Materials				14,525,000

Prices are estimates based on current Kenyan market rates from major hardware suppliers (e.g., Mace Foods, Tiles and Carpets) and quarry prices. Final costs subject to market fluctuations at time of purchase.

5.2 Equipment

Item	Description	Cost Estimate (KSh)
Excavator (Hire)	For site preparation and stone placement (60 days @ KSh 15,000/day)	900,000
Transport (Lorries)	For stone, wire mesh, and material delivery	300,000
Tools	Shovels, pliers, wire cutters, wheelbarrows, water pumps (purchase/hire)	250,000
Safety Equipment	Boots, gloves, helmets, vests for 50 workers	150,000
Total Equipment		1,600,000

5.3 Personnel

Role	No. of Persons	Rate per Day (KSh)	Total Man-Days (Est.)	Total Cost (KSh)
Project Manager	1	2,000	120	240,000
Site Supervisor/Engineer	2	1,500	120	360,000
Skilled Laborers (Assembly & Stoning)	30	700	120	2,520,000
General Laborers (Excavation & Backfill)	20	500	80	800,000
Total Personnel	53			3,920,000

6.0 BUDGET SUMMARY

Category	Total Cost (KSh)
Materials	14,525,000
Equipment	1,600,000
Personnel	3,920,000
Administrative Overhead (5% of subtotal)	1,002,250
Contingency (10% for price fluctuations/unforeseen works)	2,004,500
Total Funding Requested	23,051,750

7.0 COMMUNITY IMPACT AND SUSTAINABILITY

7.1 Community Benefits

- **Safety:** Over 5,000 residents living along the targeted sections will be directly protected from flooding and property damage.

- **Economic Stability:** Reduced infrastructure damage and business interruption will contribute to local economic stability.
- **Improved Environment:** A cleaner, more stable river corridor will enhance public health and aesthetic value for the entire community.

7.2 Skills Transfer and Training

- **On-the-Job Training:** Experienced supervisors will train the recruited youth in gabion construction techniques, stone masonry, site safety, and project management.
- **Certification:** Upon successful completion, participating youth will receive a certificate of participation, enhancing their employability for future infrastructure projects.
- **Empowerment:** The project will empower the youth by providing them with a tangible stake in the improvement of their own community, fostering a sense of ownership and pride.

7.3 Long-Term Maintenance

- **Community Stewardship Committee:** AGY will work with local community leaders to establish a "River Stewardship Committee" responsible for routine monitoring, clearing blockages, and reporting any damage to the county.
- **Annual Inspection:** AGY proposes to conduct a joint annual inspection with county officials for the first three years post-construction to ensure the structures' integrity.
- **Sensitization:** We will conduct community awareness campaigns to discourage dumping of solid waste and encroachment into the stabilized riparian reserve.

8.0 APPENDICES (Optional)

(The following documents would be attached to the hard copy of this proposal)

- **Appendix A:** Letter of Registration for Andolo Gideon Youths
- **Appendix B:** Letters of Support from Local Community Elders and Area Chief
- **Appendix C:** CVs of Key Personnel (Project Manager & Site Supervisor)
- **Appendix D:** Aerial Map/Sketch of the Proposed Project Sections
- **Appendix E:** Detailed Quotations from Material Suppliers

Conclusion

Andolo Gideon Youths is not just seeking a tender; we are seeking a partnership for a sustainable and resilient future. We have the manpower, the community trust, and the

technical commitment to deliver a high-quality, cost-effective project that will serve as a model for environmental stewardship within the county. We are ready and eager to mobilize immediately upon approval. Thank you for considering this proposal.